



**CALICUT UNIVERSITY – FOUR-YEAR UNDER
GRADUATE PROGRAMME (CU-FYUGP)**

BSc PHYSICS HONOURS

Programme	B.Sc. Physics Honours				
Course Title	ELECTRONICS I				
Type of Course	Core in Major				
Semester	II				
Academic Level	100 – 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	The course usually encompasses proficiency in mathematics, physics, and basic circuit theory, alongside computer literacy and potentially some laboratory experience, ensuring students have the foundational knowledge needed for the course material.				
Course Summary	The course provides students with a comprehensive introduction to fundamental concepts in electronics, including circuit analysis, semiconductor devices and digital logic, equipping them with the essential skills and knowledge needed to understand and work with electronic systems.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Define the basic concepts of semiconductor physics, including energy bands, charge carriers, and doping.	Remember	Definitions and basic concepts	Quizzes
CO2	Explain the operating principles of semiconductor diodes, including forward and reverse bias conditions.	Understand	Laws and theories of semiconductor physics	Problem sets, concept maps
CO3	Analyse the applications of semiconductor diodes in rectification, clipping, and clamping circuits.	Analyse	Semiconductor device applications	Research papers, case studies
CO4	Explain the principles of operation of bipolar junction transistors (BJTs) and field-effect transistors (FETs), including their modes of operation and characteristics.	Understand	Laws and theories of semiconductor physics	Problem sets, concept maps
CO5	Apply transistor models to analyse amplifier circuits.	Apply	Application of principles	Laboratory experiments, simulations
CO6	Define the basic concepts of digital electronics, including binary number systems, hexadecimal number systems	Remember	Definitions and basic concepts	Quizzes
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	SEMICONDUCTOR PHYSICS		10	15
	1	Semiconductor	3	
	2	P N Junction	2	
	3	Break down and Knee Voltage	2	
	4	Special Purpose diodes: LED and Photodiode	3	
	Sections : 5.1 – 5.19, 7.2 – 7.10 of Book 1			
II	SEMICONDUCTOR DIODE APPLICATIONS		10	20
	5	Rectifier	2	
	6	Half Wave and Full Wave Rectifiers	2	
	7	Filter Circuits	2	
	8	Voltage Multipliers	2	
	9	Zener Diode as Voltage Stabiliser	2	
	Sections : 6.8 – 6.28, Book 1			
III	TRANSISTORS		15	25
	10	Transistor	2	
	11	Transistor Connections	3	
	12	Transistor as an amplifier	3	
	13	Faithful Amplification and transistor biasing	3	

	14	Methods of Transistor Biasing – Base resistor method, Voltage Divider method, Design of transistor biasing circuits	4	
	Sections: 8.1 – 8.14, 8.16 – 8.23, 9.1 – 9.16, 9.18, Book 1			
IV	DIGITAL FUNDAMENTALS		10	
	15	Analog and Digital Signals	2	10
	16	Binary Number System	2	
	17	Decimal to Binary Conversion	2	
	18	Hexadecimal Number System	2	
	19	Binary-Coded Decimal Code	2	
	Sections: 26.1 – 26.6, 26.8 – 26.9, Book 1			
V	PRACTICALS		30	
	Conduct any 6 experiments from the given list and 1 additional experiment, decided by the teacher-in-charge, related to the content of the course. The 7 th experiment may also be selected from the given list. Other experiments listed here may be used as demonstrations of the concepts taught in the course.			
	1	Study the V-I characteristics of diodes. - Characteristics of Ge, Si diodes, LEDs and photodiode. - Reverse characteristics - Germanium diode; AO79 may give better results. - ExpEYES may be used. https://expeyes.in/experiments/electronics/diodeIV.html - Optional: Plot and fit the experimental data with the diode equation in GeoGebra or any other application and calculate the value of the ideality factor of the PN junction.		
	2	Study the characteristics of Zener diode and construct a voltage regulator. - Study the V-I characteristics of zener diode and hence determine the breakdown voltage. https://expeyes.in/experiments/electronics/zenerIV.html - Construct a voltage regulator using a zener diode and determine the percentage of voltage regulation.		

	3	Study the V-I characteristics of solar cell and find the open circuit voltage, short circuit current and maximum power point. - Plot the V-I characteristics of solar cell under dark and illuminated conditions and get the open circuit voltage and short circuit current. - Plot voltage-power graph and get the maximum output power point. - Optional: find the efficiency of the solar cell, if a standardized light source is available. - ExpEYES may be used. Solar cell of voltage rating 3V and current rating of the order of 100mA is desirable for the study. https://expeyes.in/experiments/electronics/diodeIV.html		
	4	Construction of the Half Wave Rectifier. - Construct a half wave rectifier. Breadboard may be used for the easy replacement of the filters. - Observe the waveforms without filter and with filter capacitors of four different values (4.7uF, 10uF, 47uF, 100uF) using CRO/ExpEYES. Measure the voltages and calculate the ripple factor. - Observe the variation of the ripple factor when filters of different values are used, by maintaining a low value of the load resistance.		
	5	Construction of the center tapped full wave rectifier and regulated power supply. - Construct a center tapped full wave rectifier without filter and with a filter. - Connections may be realized through soldering, to get an experience of soldering. - Measure the AC and DC voltages using a multimeter and calculate the ripple factor without and with a filter. - Observe the variation of the ripple factor with load resistance, when filter is used. - Construct 5V/12V regulated power supply using 78XX IC.		
	6	Construction of the Bridge rectifier. - Construct a bridge rectifier. Breadboard may be used for the easy replacement of the filters. - Observe the waveforms without filter and with filter capacitors of four different values (4.7uF, 10uF, 47uF, 100uF) using CRO. Measure the voltages and calculate the ripple factor. - Observe the variation of the ripple factor when filters of different values are used, by maintaining a low value of the load resistance.		
	7	Realize clipping and clamping circuits using diodes and observe the waveforms.		

		- Construct circuits using ordinary and zener diodes to clip the top, or bottom, or both of a waveform at a particular dc level. - Construct positive and negative clamper circuits and analyse the waveforms using CRO/ExpEYES. https://expeyes.in/experiments/electronics/clipping.html https://expeyes.in/experiments/electronics/clamping.html		
	8	Transistor input, output & transfer characteristics in CE configuration. - Draw the static characteristics of the transistor in common emitter configuration and calculate input/output resistance and the current gain. - ExpEYES may be used https://expeyes.in/experiments/electronics/npn.html		
	9	Construction of voltage multiplier (Doubler and Tripler). Construct the voltage doubler and tripler using diodes and capacitors and study the variation of ripple factor with respect to the capacitance values.		
	10	Study the characteristics of LDR. - Measure the dark resistance of LDR - Place LDR at different distances from an electric lamp and measure its resistance. Plot light intensity ($E \propto \frac{1}{r^2}$) vs LDR resistance. - Optional: Construct a dark sensor using LDR and transistor. In order to turn on the LED in the desired light intensity, an adjustable resistor can be used in the circuit.		

Books and References:

- Principles of Electronics by V K Mehtha and Rohith Mehtha (Book 1)
- Electronics lab manual by K A Navas (vol 1 & 2)
- Electronic Devices and Circuit Theory by Robert L. Boylestad and Louis Nashelsky
- Electronic Principles by Albert Malvino and David J. Bates
- Analog Electronics: Devices, Circuits, and Techniques by Chitrlekha Mahanta
- Basic Electrical and Electronics Engineering by R.K. Rajput
- Semiconductor Devices: Physics and Technology by S. M. Sze

.Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PSO5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	1	0	0	3	0	1	0	0	1	0	0	0

CO 2	3	1	0	0	3	0	1	0	0	1	0	0	0
CO 3	3	2	3	0	3	0	1	0	1	1	2	0	0
CO 4	3	1	3	0	3	0	1	0	0	1	0	0	0
CO 5	3	2	3	1	2	1	1	0	2	2	3	0	0
CO 6	3	1	0	0	2	1	1	0	0	1	0	0	0

Correlation Levels:

Level	Correlation
0	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Discussion / Seminar
- Internal Theory / Practical Exam
- Assignments / Viva
- End Semester Exam (70%)

Mapping of COs to Assessment Rubrics

	Internal Theory / Practical Exam	Assignment / Viva	Practical Skill Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓	✓		✓
CO 4	✓	✓		✓
CO 5	✓	✓		✓
CO 6		✓	✓	